University of Illinois
ECE 310 **Profs. Do, Snyder, Moustakides**

Midterm Exam 1

7:00-9:00pm, Wednesday, September 27, 2023

Name: _____

Section: 9:00 AM 12:00 PM 3:00 PM

NetID: _____

Score: _____

Problem	Pts.	Score
1	12	
2	12	
3	12	
4	5	
5	9	
6	20	
7	20	
8	10	
Total	100	

Instructions

- You may not use any books, calculators, or notes other than one handwritten two-sided sheet of 8.5" x 11" paper.
- Show all your work to receive full credit for your answers.
- When you are asked to "calculate", "determine", or "find", this means providing closed-form expressions (i.e., without summation or integration signs).
- Neatness counts. If we are unable to read your work, we cannot grade it.
- Turn in your entire booklet once you are finished. No extra booklet or papers will be considered.

Netid: _____

(12 Pts.)

1. Select **True** or **False** to each of the following statements:

   (a) Any causal discrete-time system must also be time-invariant. **True/False**

   (b) The ROC of a given z-transform cannot contain any poles or zeros. **True/False**

   (c) Two BIBO stable systems connected in parallel always form a BIBO stable system. **True/False**

   (d) For a discrete-time system with impulse response $h[n]$, the system output $y[n]$ to any input signal $x[n]$ is always given by $y[n] = x[n] * h[n]$. **True/False**

   (e) Two LTI systems given by impulse responses $h_1[n]$ and $h_2[n]$ are connected in series in some order, i.e. $h_1[n]$ or $h_2[n]$ may come first. If we pass an input signal $x[n]$ to this system, we will receive the same output signal $y[n]$ for either ordering of $h_1[n]$ and $h_2[n]$. **True/False**

   (f) A BIBO stable LTI system with transfer function $H(z) = \frac{1}{1-3z^{-1}}$ must be non-causal. **True/False**

(12 Pts.)

2. For each of the systems with input $x[n]$ and output $y[n]$ shown in the table, indicate by "**yes**" or "**no**" whether the properties indicated apply to the system. You will only be graded on your answers in the boxes and not on any work you show.

	Linear	Time-invariant	Causal	Stable
$y[n] = \log(\lvert n \rvert + 1)x[n]$				
$y[n] = x[n] * u[n+1]$				
$y[n] = x[n] + 3$				

Netid: _____

(12 Pts.)

3. For each of the following parts, calculate the convolution $y[n] = x[n] * h[n]$.

   (a) $\{x[n]\}_{n=0}^{4} = \{\underset{\uparrow}{1}, 2, 3, 2, 1\}, \quad \{h[n]\}_{n=0}^{1} = \{\underset{\uparrow}{-1}, 1\}.$

   (b) $x[n] = \left(\frac{3}{4}\right)^n u[n], \quad h[n] = 2u[n] - u[n-1] - u[n-2]$

Netid: _____

(5 Pts.)

4. Suppose we have an LTI system described by impulse response $h[n]$. We know that the system response to input $x[n] = u[n]$ is given by $y[n] = \delta[n] + \delta[n-1]$. Which of the following sequences correctly expresses $h[n]$? **(Circle one)**

   (a) $h[n] = \{\underset{\uparrow}{1}, 0, 1\}$

   (b) $h[n] = \{\underset{\uparrow}{1}, 0, -1\}$

   (c) $h[n] = \{\underset{\uparrow}{1}, 1\}$

   (d) $h[n] = \{\underset{\uparrow}{1}, -1\}$

(9 Pts.)

5. Calculate the z-transform $X(z)$ of $x[n] = nu[n+1]$.

Netid: _____

(20 Pts.)

6. We are given the following transfer function $H(z)$ of an LTI system

$$H(z) = \frac{1 - z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - 2z^{-1})}.$$

(a) Determine all possible ROCs and the corresponding impulse response $h[n]$ for each ROC.

(b) Determine the corresponding difference equation with transfer function $H(z)$.

Netid: _____

(20 Pts.)

7. Consider a **causal** LTI system with the following transfer function:

$$H(z) = \frac{1 - 3z^{-1} + 2z^{-2}}{1 + \frac{3}{4}z^{-1} - \frac{1}{4}z^{-2}}.$$

(a) Determine the poles, zeros, and ROC of the system.

Poles: _____

Zeros: _____

ROC: _____

(b) Calculate the system response $y[n]$ to input signal $x[n] = u[n]$.

(c) Is the system described by $H(z)$ BIBO stable? If the system is not BIBO stable, give an example of a bounded input that will produce an unbounded output.

Netid: _____

(10 Pts.)

8. Suppose we have an LTI system described by

$$y[n] - \frac{2}{3}y[n-1] - \frac{8}{9}y[n-2] = 3x[n] + \alpha x[n-1],$$

where α is a finite, real-valued constant and there exists a transfer function $H(z)$ for the system.

(a) Assuming the system is **causal**, for what value(s) of α is this system BIBO stable?

(b) Assuming the system is **non-causal** and **two-sided**, for what value(s) of α is this **non-causal** system BIBO stable?